

Chapter II

DISCRETE RANDOM VARIABLES - Part II

1 Functions of Random Variables

Let X be a discrete r.v., with PMF given by

$$P_X(x) = \begin{cases} p_i, & x = x_i \\ 0, & \text{otherwise} \end{cases}, \quad 0 < p_i < 1, p_1 + p_2 + \dots + p_n = 1,$$

D_X the range of P , and $g : D_X \rightarrow \mathbb{R}$ a real, bijective function.

Then, the PMF of the r.v. $Y = g(X)$ is:

$$P(Y = y) = P(g(X) = y) = P(X = g^{-1}(\{y\})) = P(X = x) = p,$$

therefore the PMF of r.v. Y is:

$$P_Y(y) = \begin{cases} p_i, & y = g(x_i) \\ 0, & \text{otherwise} \end{cases}, \quad 0 < p_i < 1, p_1 + p_2 + \dots + p_n = 1.$$

Example The amplitude V of a sinusoidal signal is a r.v. with PMF

$$P_V(x) = \begin{cases} \frac{1}{7}, & x = -3, -2, \dots, 2, 3 \\ 0, & \text{otherwise.} \end{cases}$$

Let $Y = \frac{V^2}{2}$ be the average power of the transmitted signal. Find the PMF of V .

The possible values of Y are 0, 0.5, 2, 4.5, so the PMF of Y is:

$$P_Y(x) = \begin{cases} \frac{1}{7}, & x = 0 \\ \frac{2}{7}, & x = 0.5, 2, 4.5 \\ 0, & \text{otherwise.} \end{cases}$$

Consider now two discrete r.v. X and Y , D_X the range of X , D_Y the range of Y . Then

$$h : D_X \times D_Y \rightarrow \mathbb{R}, h(x_i, y_j) = \overset{not}{=} q_{ij}$$

with

$$P(h(X, Y) = q) = P((X, Y) \in h^{-1}(q)).$$

2 Averages. Variance and Standard Deviation

A mode of a discrete r.v. X is a number x_{mod} satisfying

$$P_X(x_{mod}) \geq P_X(x),$$

for each value of x .

A median of a discrete r.v. X is a number x_{med} that satisfies

$$P(X < x_{med}) = P(X > x_{med}).$$

The expected value of X is

$$E(X) = \sum_{x \in S_X} x \cdot P_X(x).$$

Remark 2.1. Neither the mode nor the median of r.v. X need to be unique. A r.v. can have several modes or medians.

Example 1. For one quiz, ten students have the following grades:

$$9, 5, 10, 8, 4, 7, 5, 5, 8, 7.$$

Find the mean, the median and the mode.

The mean (*media aritmetică*) is:

$$m = \frac{9 + 5 \cdot 3 + 10 + 2 \cdot 8 + 4 + 3 \cdot 5 + 2 \cdot 7}{10} = 6.8$$

The median is 7 since are four scores below 7 and four score above 7.

The mode is 5 since it occurs more often than any other.

2. Determine the expected value of r.v. X whose PMF is

$$P_X(x) = \begin{cases} \frac{1}{8}, & x = 0 \\ \frac{3}{8}, & x = 1 \\ \frac{3}{8}, & x = 2 \\ \frac{1}{8}, & x = 3 \\ 0, & \text{otherwise.} \end{cases}$$

The expected value (*valoarea medie*) of r.v. X is:

$$E(X) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{3}{2}.$$

Proposition 2.1. For any r.v. X , the following properties hold:

1. $E(a \cdot X + b) = a \cdot E(X) + b, \quad \forall a, b \in \mathbb{R}.$
2. $E(X - E(X)) = 0;$
3. $E(X + Y) = E(X) + E(Y).$

Proposition 2.2. 1. The Bernoulli (p) r.v. X has the expected value $E(X) = p.$

2. The geometric (p) r.v. X has the expected value $E(X) = \frac{1}{p}.$

3. The Poisson (α) r.v. X has the expected value $E(X) = \alpha.$

4. The binomial (n, p) r.v. X has the expected value $E(X) = n \cdot p.$

5. The Pascal (k, p) r.v. X has the expected value $E(X) = \frac{k}{p}.$

6. The discrete (k, l) r.v. X has the expected value $E(X) = \frac{k+l}{2}.$

The variance of r.v. X (*dispersia*) is the number given by

$$\text{var}(X) = E[(X - E(X))^2].$$

The standard deviation of r.v. X (*abaterea medie pătratică*) is

$$\sigma_X = \sqrt{\text{var}(X)}.$$

Remark 2.2. We think of sample values within σ_X of the expected values $x \in [E(X) - \sigma_X, E(X) + \sigma_X]$ as "typical" values of X and other values as "unusual."

Proposition 2.3. 1. For any r.v. X , the variance can be computed using the formula:

$$\text{var}(X) = E(X^2) - E(X)^2.$$

2. The variance of any r.v. X is a positive number:

$$\text{var}(X) > 0.$$

3.

$$\text{var}(a \cdot X) = a^2 \cdot \text{var}(X), \quad \forall a \in \mathbb{R}.$$

Proposition 2.4. 1. The Bernoulli (p) r.v. X has the variance $\text{var}(X) = p(1 - p)$.

2. The geometric (p) r.v. X has the variance $\text{var}(X) = \frac{1 - p}{p^2}$.

3. The Poisson (α) r.v. X has the variance $\text{var}(X) = \alpha$.

4. The binomial (n, p) r.v. X has the variance $\text{var}(X) = np(1 - p)$.

5. The Pascal (k, p) r.v. X has the variance $\text{var}(X) = \frac{k(1 - p)}{p^2}$.

6. The discrete (k, l) r.v. X has the variance $\text{var}(X) = \frac{(l - k)(l - k + 2)}{12}$.

The n-th moment for r.v. X is $E(X^n)$.

The n-th central moment for r.v. X is $E[(X - E(X))^n]$.

Example In an experiment to monitor two calls, the PMF of X – the number of voice calls, is

$$P_X(x) = \begin{cases} \frac{1}{10}, & x = 0 \\ \frac{2}{5}, & x = 1 \\ \frac{1}{2}, & x = 2 \\ 0, & \text{otherwise.} \end{cases}$$

Find:

- a) The expected value of X ;
- b) The expected value of X^2 ;
- c) The variance of X .
- d) The standard deviation for X .

Solution: a) $E(X) = 0 \cdot \frac{1}{10} + 1 \cdot \frac{2}{5} + 2 \cdot \frac{1}{2} = \frac{14}{10}$.

b) $E(X^2) = 0^2 \cdot \frac{1}{10} + 1^2 \cdot \frac{2}{5} + 2^2 \cdot \frac{1}{2} = \frac{24}{10}$.

c) $var(X) = E(X^2) - E(X)^2 = \frac{24}{10} - \left(\frac{14}{10}\right)^2 = 0.44$

d) $\sigma(X) = \sqrt{var(X)} = \sqrt{0.44}$.

3 Conditional Probability Mass Function

Given an event B , $P(B) > 0$, the **conditional PMF** of X is:

$$P_{X|B}(x) = P(X = x \mid B).$$

Proposition 3.1. If $B_1, B_2, \dots, B_m$ is an event space, then:

$$P_X(x) = \sum_{i=1}^m P_{X|B_i} \cdot P(B_i).$$

Proposition 3.2. When a conditioning event B is included in the range of X , then the conditional PMF of X given B is:

$$P_{X|B}(x) = \frac{P(X = x | B)}{P(B)} = \begin{cases} \frac{P_X(x)}{P(B)}, & x \in B \\ 0, & \text{otherwise} \end{cases}$$

Example The length of a fax is a discrete r.v. which has PMF

$$P_X(x) = \begin{cases} 0.15, & x = 1, 2, 3, 4 \\ 0.1, & x = 5, 6, 7, 8 \\ 0, & \text{otherwise} \end{cases}$$

Suppose the company has two fax machines, one for shorter than 5 pages and the other for faxes that have more pages.

- a) What is the PMF of fax length in the second machine?
- b) Find the expected value, the variance and the standard deviation of $X | B$.

Solution a) The condition is $B = \{5, 6, 7, 8\}$, so the probability of B is $P(B) = 4 \cdot 0.1 = 0.4$. The conditional PMF of X given B is:

$$P_{X|B}(x) = \begin{cases} \frac{P_X(x)}{P(B)}, & x \in B \\ 0, & \text{otherwise} \end{cases} = \begin{cases} \frac{0.1}{0.4}, & x \in B \\ 0, & \text{otherwise} \end{cases} = \begin{cases} \frac{1}{4}, & x \in \{5, 6, 7, 8\} \\ 0, & \text{otherwise} \end{cases}$$

$$\text{b) } E(X | B) = 5 \cdot \frac{0.1}{0.4} + 6 \cdot \frac{0.1}{0.4} + 7 \cdot \frac{0.1}{0.4} + 8 \cdot \frac{0.1}{0.4} = 26 \cdot 0.25 = 6.5 \text{ pages.}$$

Now,

$$\text{var}(X | B) = E(X^2 | B) - [E(X | B)]^2.$$

First, we compute

$$E(X^2 | B) = 25 \cdot 0.25 + 36 \cdot 0.25 + 49 \cdot 0.25 + 64 \cdot 0.25 = 174 \cdot 0.25 = 43.5 \text{ pages, so}$$

$$\text{var}(X | B) = 43.5 - 6.5^2 = 1.25 \text{ pages.}$$

4 Solved Problems

1. A student takes two courses. In each course, the student will earn a B with probability 0.6, or a C with probability 0.4, independent to the other course.

To calculate a grade point average, a B worth 3 points and a C worth 2 points. The student's average is the sum of the points for each course divided by 2.

Make a table of a simple space of the experiment and the corresponding values of the student's average, X . Write the corresponding cumulative distribution function (CDF).

Solution

$X :$	BB	BC	CB	CC
	3	$\frac{5}{2}$	$\frac{5}{2}$	2

The corresponding probabilities are: $P(X = 3) = 0.6 \cdot 0.6 = 0.36$, $P(X = \frac{5}{2}) = 2 \cdot 0.6 \cdot 0.4 = 0.48$, $P(X = 2) = 0.4 \cdot 0.4 = 0.16$ therefore, the PMF for X given B is:

$$P_X(x) = \begin{cases} 0.36, & x = 3 \\ 0.48, & x = \frac{5}{2} \\ 0.16, & x = 2 \\ 0, & \text{otherwise.} \end{cases}$$

The corresponding cumulative distribution function is given by:

$$F_X(x) = \begin{cases} 0, & x < 3.36, \quad x \in [3, \frac{5}{2}) \\ 0.84, & x \in [\frac{5}{2}, 2) \\ 1, & x \in [2, \infty). \end{cases}$$

2. The manager of a data processing center reports, during the previous 100 days, the information system had 3 failures.
 - a) What is the probability that, given a day, there are no failures?
 - b) What is the probability of one or more failures on a given day?
 - c) What is the probability of at least 2 failures in 3 days?

Solution a) The r.v. X is a Poisson r.v. with $\lambda = \frac{3}{100} = 0.03$, so its PMF is:

$$P_X(x) = \begin{cases} \frac{e^{-0.03}(0.03)^x}{x!}, & x = 0, 1, 2, \dots \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, $P(X = 0) = e^{-0.03} = 0.97$.

b) $P(X = 0 \cup X = 1 \cup X = 3) = 1 - P(X = 0) = 1 - 0.97 = 0.03$.

c) The corresponding value of λ for 3 days is $\lambda = \frac{9}{100} = 0.09$, therefore

$$P(X \geq 2) = 1 - P(X \leq 1) = 1 - P(X = 0) - P(X = 1) = 1 - 0.91 - 0.08 = 0.003.$$

3. Your company receives a batch of 20 articles. Since it is very expensive to control them all, you have adopted the strategy of checking a random sample of 6 and not proceeding with the check if the most one is found to be defective.

What is the probability that 5 defective items do not undergo additional checks?

Solution $P(X = 0 \cup X = 1) = P(X = 0) + P(X = 1) = \frac{C_5^0 \cdot C_{15}^6}{C_{20}^6} + \frac{C_5^1 \cdot C_{15}^5}{C_{20}^6}$.

4. At the beginning of August, the heads of an American college found that they could admit some more students in addition to the maximum number fixed: the additional registrations would not increase the expenses, while on the other hand they would increase the income.

From the previous experience, they know that only 40% of the admitted students actually will participate at the lectures.

- a) What is the probability that a maximum of 6 students will enroll if the number of students is increased by 10 students?
- b) If 70% of admitted students will enroll, what is the probability that

at least 12 out of 15 students enroll?

Solution a) For $n = 10$ and $p = 0.4$ the probability we asked for is:

$$\begin{aligned} P(X \leq 6) &= 1 - P(X = 7) - P(X = 8) - P(X = 9) - P(X = 10) \\ &= 1 - C_{10}^7(0.4)^7(0.6)^3 - C_{10}^8(0.4)^8(0.6)^2 - C_{10}^9(0.4)^9(0.6)^1 - C_{10}^{10}(0.4)^{10} \\ &= 1 - 0.0548 = 0.9452. \end{aligned}$$

b) For $n = 115$ and $p = 0.7$ the probability is:

$$\begin{aligned} P(X \geq 12) &= P(X = 12) + P(X = 13) + P(X = 14) + P(X = 15) \\ &= C_{15}^{12}(0.7)^{12}(0.3)^3 + C_{15}^{13}(0.7)^{13}(0.3)^2 + C_{15}^{14}(0.7)^{14}(0.3)^1 + C_{15}^{15}(0.7)^{15}. \end{aligned}$$

5. Let X denote the number of additional years that a randomly chosen 70 year old person will live. If the person has high blood pressure, denoted as event H , then X is a geometric ($p = 0.1$) r.v.

Otherwise, if the person's blood pressure is regular, event R , then X has geometric ($p = 0.05$) PMF.

- a) Find the conditional probability mass functions $P_{X|H}(x)$ and $P_{X|R}(x)$.
b) If 40% of all 70 years old have high blood pressure, what is the PMF of X ?

Solution a)

$$P_{X|H}(x) = \begin{cases} 0.1 \cdot (0.9)^{x-1}, & x = 1, 2, \dots \\ 0, & \text{otherwise.} \end{cases}$$

and, respectively:

$$P_{X|R}(x) = \begin{cases} 0.05 \cdot (0.95)^{x-1}, & x = 1, 2, \dots \\ 0, & \text{otherwise.} \end{cases}$$

- b) Due to the fact that $\{H, R\}$ is an event space, using Law of Total Probability, we obtain:

$$P_X(x) = P_{X|H}(x) \cdot P(H) + P_{X|R}(x) \cdot P(R)$$

$$= \begin{cases} 0.4 \cdot 0.1 \cdot (0.9)^{x-1} + 0.6 \cdot 0.05 \cdot (0.95)^{x-1}, & x = 1, 2, \dots \\ 0, & \text{otherwise.} \end{cases}$$

6. The r.v. X is the number of pages in a facsimile transmission. Based on experience, you have a probability model $P_X(x)$ for the number of pages in each fax you send.

The phone company offers you a new charging plan for faxes: 0.10 for the first page, 0.09 for the second one,..., 0.06 for the 5th page. For all faxes between 6 and 10 pages, the phone company will charge 0.5 per fax (faxes longer than 10 pages are not accepted).

- Find a function $g(X) = Y$ for the charge in cents for sending a fax.
- Suppose all your faxes contain 1, 2, 3 or 4 pages with equal probability. Find PMF and the expected value of Y , the charge for a fax.
- If the probability model of the number of pages X is

$$P_X(x) = \begin{cases} 0.15, & x = 1, 2, 3, 4 \\ 0.1, & x = 5, 6, 7, 8 \\ 0, & \text{otherwise,} \end{cases}$$

and suppose that the firm sends all faxes with an even number of pages to fax machine A and the rest to fax machine B , then find the PMF of the length X of a fax, given the fax was sent to A . What are the conditional expected length and standard deviation?

- Find the conditional PMF of the length X of a fax, given the fax was sent to B and have no more than 6 pages. What are the conditional expected value and standard deviation?

Solution a) Consider the function $g(x) = ax^2 + bx + c$, and find the coefficients a, b, c such that $g(1) = 10, g(2) = 19, g(3) = 27$. After a straightforward computation, we find $a = 10.5, b = 0.5, c = 0$, therefore

$$Y = g(X) = \begin{cases} 10.5X^2 - 0.5X, & X = 1, 2, 3, 4, 5 \\ 50, & X = 6, 7, 8, 9, 10 \\ 0, & \text{otherwise.} \end{cases}$$

- The PMF of X is:

$$P_X(x) = \begin{cases} \frac{1}{4}, & x = 1, 2, 3, 4 \\ 0, & \text{otherwise} \end{cases},$$

therefore

$$P_Y(x) = \begin{cases} \frac{1}{4}, & x = 10, 19, 27, 34 \\ 0, & \text{otherwise.} \end{cases},$$

and the expected value of Y is:

$$E(Y) = \frac{1}{4}(10 + 19 + 27 + 34) = 22.5 \text{ cents.}$$

c)

$$P_{X|A}(x) = \begin{cases} \frac{P_X(x)}{P(A)}, & x \in A \\ 0, & \text{otherwise} \end{cases} = \begin{cases} \frac{0.15}{0.5}, & x = 2, 4 \\ \frac{0.1}{0.5}, & x = 6, 8 \\ 0, & \text{otherwise} \end{cases}$$

$$= \begin{cases} 0.3, & x = 2, 4 \\ 0.2, & x = 6, 8 \\ 0, & \text{otherwise.} \end{cases}$$

The conditional expected value of X given A is:

$$E(X | A) = 2 \cdot 0.3 + 4 \cdot 0.3 + 6 \cdot 0.2 + 8 \cdot 0.2 = 4.6,$$

and the conditional standard deviation $\sigma(X | A) = \sqrt{4.6}$.

d) Using the same arguments as for the previous question, the conditional PMF of X given B is:

$$P_{X|B}(x) = \begin{cases} 0.3, & x = 1, 3 \\ 0.2, & x = 5, 7 \\ 0, & \text{otherwise.} \end{cases}$$

The conditional PMF of X given $B \cap C$ is:

$$P_{X|B \cap C}(x) = \begin{cases} 0.3, & x = 1, 3 \\ 0.4, & x = 5 \\ 0, & \text{otherwise.} \end{cases}$$

and the corresponding conditional expected value

$$E(X | B \cap C) = 1 \cdot 0.3 + 3 \cdot 0.3 + 5 \cdot 0.4 = 3.2.$$

7. On the Internet, data is transmitted in packets. In a simple model for www traffic, the number of packets N needed to transmit a web page depends on whether the page has graphic images.
 If the page has images (event I), then N is uniformly distributed between 1 and 50 packets.
 If the page is just text (event T), then N is uniformly distributed between 1 and 5 packets.

Assuming a page has images with probability $\frac{1}{4}$, find:

- conditional probability mass functions $P_{N|I}(x), P_{N|T}(x)$;
- probability mass functions $P_N(x)$;
- conditional probability mass functions $P_{N|N \leq 3}(x)$
- conditional expected value $E[N | N \leq 3]$;
- conditional variance $var[N | N \leq 3]$.

Solution a) $k = 1, l = 50$ so

$$P_{N|I}(x) = \begin{cases} \frac{1}{50}, & x = 1, 2, \dots, 50 \\ 0, & \text{otherwise.} \end{cases}$$

$k = 1, l = 5$ so

$$P_{N|T}(x) = \begin{cases} \frac{1}{5}, & x = 1, 2, \dots, 5 \\ 0, & \text{otherwise.} \end{cases}$$

- b) The events $\{I, T\}$ are an event space, so, using the Law of Total Probabilities, we obtain:

$$P_X(x) = P_{X|I}(x) \cdot P(I) + P_{X|T}(x) \cdot P(T)$$

$$= \begin{cases} \frac{1}{50} \cdot \frac{1}{4} + \frac{1}{5} \cdot \frac{3}{4}, & x = 1, 2, \dots, 5 \\ \frac{1}{50} \cdot \frac{1}{4}, & x = 6, 7, \dots, 50 \\ 0, & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{31}{200}, & x = 1, 2, \dots, 5 \\ \frac{1}{200}, & x = 6, 7, \dots, 50 \\ 0, & \text{otherwise.} \end{cases}$$

c) The conditional PMF

$$P_{N|N \leq 3}(x) = \frac{P(N)}{P(N \leq 3)} = \frac{P(N)}{P(N=1) + P(N=2) + P(N=3)}$$

$$= \begin{cases} \frac{31}{200} \cdot \frac{200}{63}, & x = 1, 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{3}, & x = 1, 2, 3 \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{d) } E[N | N \leq 3] = 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{3} = 2.$$

$$\text{e) } \text{var}[N | N \leq 3] = E[N^2 | N \leq 3] - (E[N | N \leq 3])^2.$$